

SAMAYUKTAM

SPCMP v1 · Cross-Hardware Verification Report

Profile: 0x01 · n = 10,000 float32 values · Seed: 42

1. Executive Summary

SPCMP v1 (Samayuktam Profile for PCMP v1) was executed across four heterogeneous hardware targets: Google CPU (x86-64 AVX2), Google TPU (BF16 systolic), Google GPU (CUDA/XLA), and a Native GPU (local CUDA driver). All four nodes produced identical cryptographic artifacts after preprocessing canonicalization. Without SPCMP, each node produced a divergent hash due to hardware-dependent NaN payloads and signed-zero representation differences.

Metric	Without SPCMP	With SPCMP
Hash consensus	✗ Divergent	✓ Unified
Diverged floats	30 (NaN×20 + -0×10)	0
Merkle proof	N/A	✓ Valid — all 4 nodes
Compression ratio	—	1.545×

2. Hash Comparison: Raw vs SPCMP

The table below shows the flat SHA-256 hash (first 4 bytes) for each hardware target, in both raw (non-canonicalized) and SPCMP-processed modes.

Hardware	Arch	Raw Hash	SPCMP Hash	Verify
Google CPU	x86-64 AVX2	0x336D3BDC	0x52FC93DB	✓
Google TPU	BF16 systolic	0x3B86E4A4	0x52FC93DB	✓
Google GPU	CUDA / XLA	0xF49C104C	0x52FC93DB	✓
Native GPU	Local CUDA driver	0x77BBC419	0x52FC93DB	✓

3. Per-Node Run Details

3.1 Google CPU (x86-64 AVX2)

Profile: spcmp_v1 · 0x01

Flat hash: 0x52FC93DB

Merkle root: 2ff6d95f304ec058...

NaN canonicalized: 20

−0 canonicalized: 10

Compressed bytes: 25,884

Compression ratio: 1.545×

Elapsed: 13.218 ms

Verification: PASS

3.2 Google TPU (BF16 systolic)

Profile: spcmp_v1 · 0x01

Flat hash: 0x52FC93DB

Merkle root: 2ff6d95f304ec058...

NaN canonicalized: 20

−0 canonicalized: 10

Compressed bytes: 25,884

Compression ratio: 1.545×

Elapsed: 13.831 ms

Verification: PASS

3.3 Google GPU (CUDA / XLA)

Profile: spcmp_v1 · 0x01

Flat hash: 0x52FC93DB

Merkle root: 2ff6d95f304ec058...

NaN canonicalized: 20

−0 canonicalized: 10

Compressed bytes: 25,884

Compression ratio: 1.545×

Elapsed: 12.657 ms

Verification: PASS

3.4 Native GPU (Local CUDA driver)

Profile: spcmp_v1 · 0x01

Flat hash: 0x52FC93DB

Merkle root: 2ff6d95f304ec058...

NaN canonicalized: 20

−0 canonicalized: 10

Compressed bytes: 25,884

Compression ratio: 1.545×
Elapsed: 13.725 ms
Verification: PASS

4. Shared Merkle Root

All four nodes independently produced the identical Merkle root over the compressed float sequence:

2ff6d95f304ec058ace948755de7dfbb0759b1e8d7185b2df719b2ba350162e1

The Merkle tree is constructed over 4096-byte chunks of the zlib-compressed delta stream. Its root constitutes the cryptographic proof of canonical equivalence across hardware targets.

5. Preprocessing Canonicalization (SPCMP §4)

Prior to PCMP v1 canonical ordering, every float32 element is passed through the SPCMP preprocessing gate exactly once:

Condition	Output	Rationale
Any NaN (any payload)	0x7FC00000	Eliminates hardware-dependent NaN payload divergence
−0.0 (0x80000000)	0x00000000	Unifies signed-zero representation across FPUs
All other values	Preserved	Bit-exact pass-through

The mapping is deterministic, idempotent, and explicitly non-invertible outside the Canonical Value Domain (CVD). Loss of NaN payload and signed-zero sign bit is intentional and normative.

6. Test Parameters

Float count (n): 10,000
Random seed: 42
NaN variants injected: 20 (payloads: 0x7FC00001, 0x7FC0DEAD, 0xFFFFFFFF, 0x7F800001, 0xFFC00000)
−0.0 injected: 10
Total diverged: 30
SPCMP profile: spcmp_v1 · 0x01
Compression: zlib level 9 · little-endian uint32
Hash algorithm: SHA-256 (Merkle over 4096-byte chunks + flat)